

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

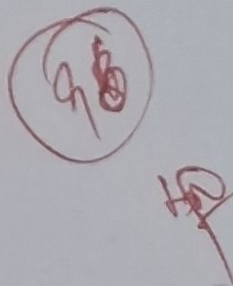
Descriptive Written Test

Code of Conduct:

I Preethi R (192511352) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. Preethi



Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.
3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

CO-1 ASSESSMENT TOOL - 8

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Course code: CSA1710

Course name: Artificial Intelligence

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DESCRIPTIVE TEST

Smart Home application automation System

Introduction:

- * A Smart home automation System uses intelligent agents to control lighting, temperature and security based on sensor inputs such as motion detection, temperature and time.
- * It also learns user preference to improve comfort and energy efficiency.

Simple Reflex agent

A Simple reflex agent makes decision based only on the current input without considering previous events.

Working in the Smart Home:

- Turns on light when motion is detected.
- Turns off light when no motion is detected.
- Switches on their air conditioner if the temperature exceeds a predefined value.
- Sounds an alarm immediately when an intruder is detected.

Advantages:

- Fast response
- Easy to design and implement
- Suitable for simple tasks.

Disadvantage:

- Cannot remember past events
- Cannot adapt to user preference
- performs poorly in complex situations.

Model-based reflex agent

A model-based reflex agent maintains an internal model of the environment. It remembers previous states and uses them to make better decisions.

Working in the Smart Home:

- Knows whether someone is already inside a room before switching lights.
- Maintain room occupancy information
- Adjust heating or cooling based on previous temperature reading.
- Distinguish between temporary sensor errors and actual events.

Advantages:

- Makes intelligent decision
- Can plan future action
- Flexible in different situation

Disadvantages:

- * Requires planning algorithm
- * Decision-making takes more time.

Goal-based agent

A Goal-based agent choose action that help achieve a specific objective.

Working in the Smart Home:

- * Maintains a comfortable room temperature.
- * Ensure home security at night.
- * Saves electricity while keeping occupants comfortable
- * plans action such as lowering blinds during hot

afternoon.

Advantages:

- * Handles partially observable environment
- * More accurate than a simple reflex agent
- * Reduces unnecessary action

Disadvantages:

- * More complex than simple reflex agent.
- * Requires memory and additional processing

Utility-based agent

A utility based agent selects the action that provide the highest overall benefit by considering multiple factors simultaneously.

Working in the Smart Home:

- Balance comfort, security, energy consumption
- learns user preference over time.

→ Adjust room temperature according to user habit
→ choose setting that reduce electricity bills while maintaining comfort.

Advantages:

- * produce optimal decision
- * Learn and adapt over time.
- * Balance multiple objectives efficiently.

Disadvantages:

- * Most complex to design
- * Require utility calculation and learning algorithm.

Best Agent for Smart Home:

A hybrid approach combining model-based and utility based agent is the most effective.

- * The model based agent maintain information about occupants, sensors data, and environmental condition.
- * The utility-based agent selects action that maximize to real comfort, energy efficient and security while learning user preference.

This combination enables the smart home to react intelligently to real-time event, reduce consumption of energy, improve user comfort and provide strong security.

Conclusion:

Although simple reflex and goal-based algorithm agents are useful for specific task, a hybrid model-based + utility-based agent provide the best solution because it remembers past information, learns user behaviour, optimizes resource usage and makes intelligent decision in dynamic environment.

Robot in an unknown Maze:-

Instruction:

A robot placed in an unknown maze must explore the environment and find a path to the exit without prior knowledge of the layout. Uniformed Search algorithm such as uniform Cost Search (UCS) and Iterative Deeping Search (IDS) helps the robot systematically search for the goal.

Uniform Cost Search (UCS):

Uniform Cost Search expands the node with the lowest cumulative path cost first. It guarantees the least-cost path when step costs are positive.

Working in the maze:

- * The robot starts from the initial position
- * It calculate the total cost to reach each neighboring node.
- * The path with the smallest accumulated costs is explored first.

* The process continues until the exit is found.

Advantages:

- * Finds the optimal path
- * Complete if all costs are positive
- * Suitable when movement costs differ.

Disadvantage:

- * Requires large memory
- * Can be slow for large maze.

Complexity:

Time complexity: Exponential $O(b^{1 + \lceil C^*/\epsilon \rceil})$

Space complexity: $O(n)$

Iterative Deepening Search (IDS)

IDS combine the advantages of depth-first search and breadth-first search by repeatedly increasing the depth limit.

Working on the Maze:

- * Search begins with depth limit 0.
- * If the exit is not found, the limit increases to 1, then 2 and so on.
- * The process repeats until the goal is reached.

Advantages:

- * Uses very little memory
- * Complete for finite search space
- * Finds the shallowest solution.

Disadvantage:

- * Repeat exploration of upper levels.
- * Not optimal when path costs differ.

Complexity:

Time complexity: $O(b^d)$

Space complexity: $O(b^d)$

Conclusion:

For an unknown maze, a Goal-based agent using uniform cost search provide the most intelligent solution because it plans systematically and finds the optimal path.

When memory is limited, Goal-based agent with IDS offers an efficient alternative with lower memory usage.

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand